

COURSE TERM PROJECT HANDOUT

LLM Applications in Cybersecurity: Proof of Concept (PoC) Development

1. Project Overview & Learning Objectives

This term project offers an immersive opportunity to bridge the gap between Large Language Model (LLM) theories and practical implementations within the Cybersecurity domain. Teams are expected to build, evaluate, and document a robust Proof of Concept (PoC) application addressing a critical security challenge using modern LLM techniques.

By completing this project, you will demonstrate capabilities across two core learning objectives:

- **1. Advanced Technical Application:** Successfully synthesize and deploy core techniques, model architectures, and prompt methodologies acquired throughout the course onto a meaningful, real-world scale security system.
- **2. Independent Ideation & Research:** Formulate, justify, and design a novel application scope by comprehensively consulting the foundational concepts and architectures highlighted in the **Week 12 lecture slides**

2. Core Scope & Acceptable Domains

Your project must focus on developing an intelligent application utilizing LLMs to automate, augment, or resolve specific vulnerabilities, threats, or operations within cybersecurity. Accepted implementation scopes include, but are not limited to:

- **Phishing & Social Engineering Detection:** Developing fine-tuned or prompt-engineered LLM pipelines to detect complex phishing contexts across emails, communication platforms, or SMS text channels.
- **Automated Malware & Code Vulnerability Analysis:** Using LLMs for reverse-engineering assistance, static code vulnerability hunting, malware behavior summarization, or explanatory decompilation.
- **Threat Detection & Log Auditing:** Parsing, normalizing, and correlating massive volumes of system/network logs (SIEM, firewall, EDR) to pinpoint anomalies, trace active threat actors, and extract threat intelligence indicators automatically.
- **Incident Response & Playbook Generation:** Creating conversational or autonomous orchestration agents that parse security alerts and automatically generate structured, step-by-step mitigation workflows.

3. Team Logistics & Structure

- **Team Size:** Projects may be executed individually or in collaborative teams of **1 to 4 members** maximum.

- **Accountability:** While technical components should seamlessly integrate into a unified system, individual team members are expected to maintain distinct ownership over specific subsystems or evaluation pipelines.

4. Deliverables & Milestone Timeline

To facilitate steady development, the project structured milestone checkpoints across the final weeks of the term. Submissions are split into individual track records and unified team reports:

Timeline Milestone	Submission Type	Description & Requirements
Week 14 Checkpoint	Individual Progress Report	Personal Contribution Tracking Detailed accounting of initial research, environment setup, and module prototypes. • 3+ Pages: Minimum for Grade A • 6+ Pages: Required for Grade E (Excellent)
Week 15 Checkpoint	Individual Progress Report	Personal Implementation Tracking Documentation of evaluation metrics, refining prompt engineering pipelines, fine-tuning scripts, or dataset curations. • 3+ Pages: Minimum for Grade A • 6+ Pages: Required for Grade E (Excellent)
Week 16 Final Deadline	Final Integrated Report & PoC Artifact	Shared Team Submission A single, fully unified, and identical document submitted on behalf of the whole team. Covers absolute system design, security evaluation results, and architectural limitations.

5. Incentive Structures & Grading Policy

CRITICAL POLICY: The Outstanding (O) Retroactive Upgrade

To incentivize rigorous integration and world-class engineering, any team that delivers an **Outstanding (O)** grade on their final Week 16 integrated report will trigger an automatic evaluation override. This policy retroactively replaces all individual scores obtained during the Week 14 and Week 15 checkpoints with an automatic **Outstanding (O)** mark, regardless of initial individual page counts or intermediate grades.

Immediate Action Items:

- Form engineering teams (1 to 4 members) and finalize alignment.
- Consult the Week 12 slides thoroughly to brainstorm architectural frameworks and model integration mechanisms.
- Select your cybersecurity problem domain (e.g., phishing mitigation, log processing, or binary vulnerability exploitation) and prepare development pathways ahead of the Week 14 individual check-ins.